IST 691: Research Methodology and Scientific Writing Skills

Variables Used in Research

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Outline

- Types of variables
- The independent variable
- The dependent variable

Introduction

- Variables must be identified after the research problem and hypothesis have been specified
- Variable
  - Variables in research are measurable characteristics that change or vary

Types of Variables

- Independent Variable (IV)
  - The factor the researcher changes or manipulates to observe its effect (the "cause")
- Dependent Variable (DV)
  - The outcome or effect that is measured, which changes in response to the IV (the "effect").
- Control Variable
  - Kept constant to prevent it from influencing the results, thus isolating the IV's effect
- Confounding Variable
  - An unexpected third variable that affects both the IV and DV, potentially distorting the true relationship
- Extraneous Variable
  - Any variable outside the IV that could affect the DV

Types of Variables

- Confounding vs extraneous vs control variables
  - Extraneous variables might affect results
  - Confounding variables are uncontrolled factors that influence both the IV and DV
    - Biggest threat to a study's validity because they offer an alternative explanation for the results
  - Control variables are actively managed (kept constant) to prevent them from from affecting results, ensuring only the independent variable's impact is measured

Types of Variables

- Other important classifications
  - Qualitative vs Quantitative
    - Qualitative (Categorical)
      - Variables that varies in kind
      - Non-numerical, descriptive (e.g., gender, eye color, treatment type, hair colour (Black, Brown, Gray, Red, Yellow)
  - Quantitative variables
    - Numerical (Variables that varies in amount)
      - Age, height, test scores, loudness (decibels), latency (minutes, seconds), etc.
      - Discrete: Countable, whole numbers (e.g., number of children)
      - Continuous: Can take any value within a range (e.g., height, temperature)

Types of Variables

- Other important classifications
  - Moderating Variable
    - Influences the strength or direction of the relationship between the IV and DV
    - Examples
      - Age moderates the link between exercise (IV) and weight loss (DV), making exercise more effective for younger people than older people
      - Mental health status affecting how sleep impacts grades
  - Intervening (Mediating) Variable
    - A variable that explains the process or mechanism through which the IV affects the DV
    - Examples
      - Education (IV) leads to better job skills (Mediator), which then increases income (DV), showing how education boosts earnings
      - Sleep quality (IV) affects alertness (Mediator), which influences academic success (DV)

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The Independent Variable

- In non-experimental research
  - There is often no manipulation of the independent variable (IV)
  - Instead, researchers observe and measure variables as they naturally occur, identifying potential predictors or correlates rather than causes
    - The independent variable in these studies is typically a characteristic (like age, gender) that the researcher studies for its relationship to a dependent variable, without controlling or altering it, focusing on associations, not cause-and-effect
  - Example
    - Observing how naturally occurring income relates to education, not changing income to see effects on education

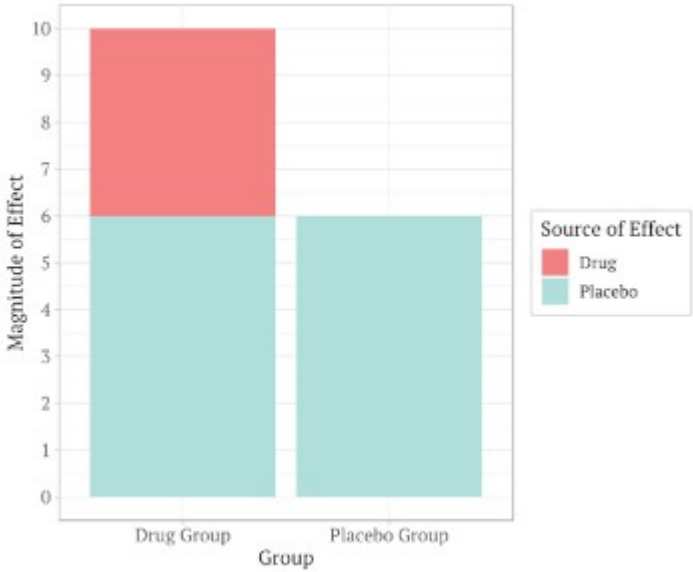
The Independent Variable

- In experimental research
  - The independent variable (IV) is the factor that a researcher deliberately changes, manipulates, or controls to observe its effect on the dependent variable (DV)
  - It's the "cause" in a cause-and-effect relationship

Variation

- Many ways in which variation in the independent variable can be achieved
  - Presence vs absence
  - Amount of variable
  - Type of variable

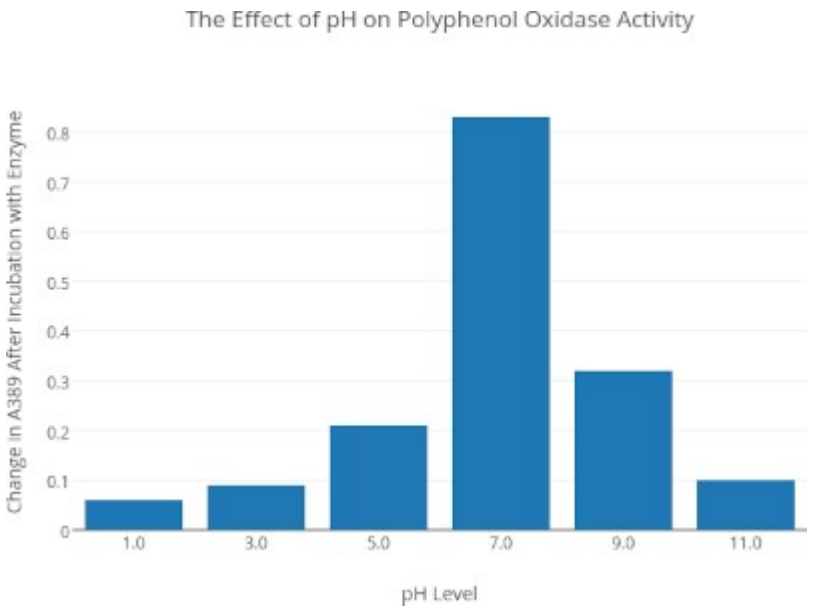
Presence vs Absence



- One group of research participants receive the treatment condition, the other group does not
- The two groups are then compared to see if the group that received the treatment condition differs from the group that did not
- Example
  - One group of research participants given a drug
  - The other group given a placebo
  - The 2 groups compared on some measure

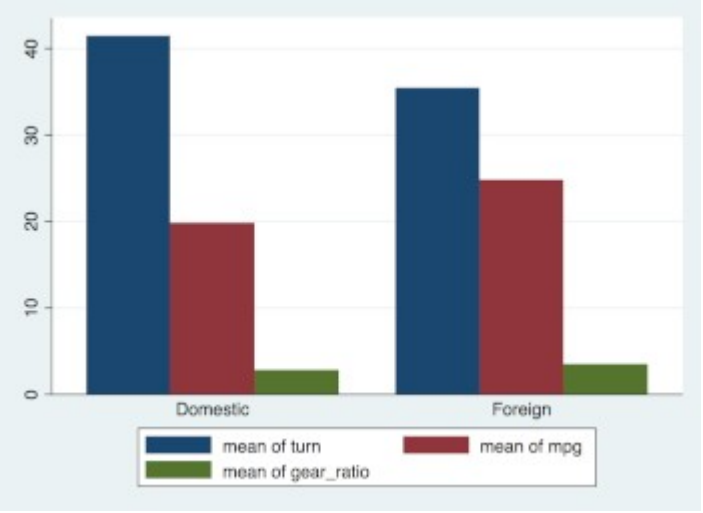
Amount of Variable

- Administer different amounts of the variable to each of several groups



Type of Variable

- Vary the kinds or categories of a variable to be compared



Establishing Variation in the Independent Variable

- It is not always easy to establish controlled variation
- 2 possible ways
  - Experimental manipulation
  - Individual difference manipulation

Experimental Manipulation

- Controlled adjustment of independent variable
- Experimenter administers
  - One specific controlled amount of a variable to one group of individuals
  - A different specific controlled amount of the same variable to a second group of individuals
  - 2 common approaches
    - Instructional manipulation
    - Event manipulation

Instructional Manipulation

- Vary the independent variable by giving a different set of instructions to to different groups of participants
  - One group of participants receives one set of instructions and another group of participants receives a different set of instructions
- Risk of the approach
  - Some participants may be inattentive when the instructions are given
  - Participant-participant variation in the interpretation of the instructions
- Minimizing the risk
  - Make instructions simple, give them emphatically, and make them relate to the activity at hand

Manipulation of Events

- Vary the independent variable by altering the events that the participants experience
- Example
  - Different dosages administered in drug research

Individual Difference Manipulation

- Vary independent variable by selecting participants that differ in the amount or type of a measured internal state (e.g., self esteem, anxiety level)
- Example
  - Select participants having different levels of a given variable (e.g., anxiety) and then look for effects of the difference

Constructing the Independent Variable

- In addition to deciding how variation in the independent variable is to be achieved, researcher must also decide how the independent variable is to be constructed (i.e., operationally defined)
- Examples
  - Specifying the exact empirical operations that define the independent variable (administer 2 mg of aspirin)
  - Display a list of words for 2 seconds
  - More complicated if the independent variable is more abstract: attitude, anxiety, learning, etc.
    - Difficulty stems from the fact that there is no one single definition or empirical referent for such constructs
    - Examining how the variable is defined in previous research may help
    - Sometimes, many techniques are available, and the difficulty is selecting which one to use in your work

Construct Validity of the Independent Variable

- Important to ask the following questions
  - Do the experimental operations to be carried out represent the abstract concept (construct) we intend it to and does it behave as theory predicts?
  - Will the different levels of variation of the independent variable make participants behave differently?
  - In what ways should they behave differently?
- This ensure that the IV isn't just a superficial measure but reflects the underlying theoretical idea, crucial for ensuring effects on the dependent variable are due to the real concept, not just a flawed operationalization or other unintended factors
- These questions are asked in order to ensure that construct validity has been attained
- Construct validity
  - The extent to which the abstract construct or conceptual variable of interest can be inferred from the operational definition of the construct

Some Guidelines for Construct Validity

- Clear definition of the abstract construct
- Then
  - Obtain data to demonstrate that the empirical representation of the independent variable produces the expected outcome
    - e.g., If rat is hungry, we expect it to consume more food (assuming we are investigating relationship between hunger and food consumption)
  - Obtain data to show that the empirical representation of the independent variable does not vary with measures of related but different conceptual variables
    - e.g., operational definition for communication expertise should be related to the level of knowledge possessed by the communicator, but not to to the communicator's power, position, trustworthiness, etc.

The Number of Independent Variables

- Sometimes one, other times more than one
- If there is more than one independent variable, we must consider interaction
- Interaction
  - The differential effect that one independent variable has for each level of one or more additional independent variables
- No theoretical limit to the number of independent variables, but realistically, there is

Limiting the Number of Independent Variables

- Research participant's point of view
  - The more the variables, the more the things to be done
    - Participating in more events, taking more tests
  - Participant is bound to become bored, irritated, or resentful, thus increasing a confounding variable in the experiment
- Researcher's point of view
  - The more the number of independent variables, the more difficult it is to make sense out of the data
  - Rule of thumb
    - The experiment should only be as complex as is necessary for the important relationships to emerge in a clear manner
      - Do not use the "why not approach" in which a variable is included because there is no real reason not to include it
      - Rather, only include those variables that seem to be important to reveal the important relationships

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The Dependent Variable

- In non-experimental research
  - The dependent variable (DV) is the outcome or response variable that is observed as it naturally changes in relation to other variables (independent or predictor variables) that are also naturally occurring
  - It's the effect or result being studied, such as customer satisfaction, stress levels, or health outcomes, in situations where direct manipulation isn't possible or ethical, unlike in experiments

The Dependent Variable

- In experimental research
  - The variable that is defined to measure the effect of the variation of the independent variable
  - Sometimes, it is hard to select a measure for the dependent variable
    - For a long time, it was assumed that split-brain surgery left the person unchanged because there seemed to be little alteration in personality or behaviour
    - But later, when the test environment was changed to assess participants' responses on a variety of specific dependent variables, it was demonstrated that split-brain surgery does have a significant impact on the person
      - Dependent variable was changed to short-term memory

The Dependent Variable

- Task of dependent variable
  - To determine whether the independent variable did or did not produce and effect
  - If an effect was produced, the dependent variable must
    - Indicate whether the effect was a facilitating one or an inhibiting one
    - Reveal the magnitude of the effect
- Researcher must determine what specific measure to use to assess the effect of the independent variable

Reliability and Validity of the Dependent Variable

- Reliability (Consistency)
  - Refers to consistency or stability
  - The extent to which the same results are obtained when responses are measured at different times under the same conditions
- Validity (Accuracy)
  - The extent to which the DV accurately reflects the concepts it is supposed to measure
- A measure can be reliable but not valid (consistent but wrong), but a valid measure is generally also reliable (accurate results are reproducible)

Reliability of the Dependent Variable

- If participants consistently respond in the same way on the independent variable, then the dependent variable is reliable
- But most experiments are conducted as single-occasion events
  - Which means that the dependent variable is observed only once
    - So, reliability of dependent variable cannot be assessed
  - However, it has been shown that reliability of the dependent variable increases as responses are aggregated over time
    - Although single-occasion experiments are unreliable, **replication** of the results of single-occasion studies provides a good measure of reliability of the dependent measures
      - Because replication of an effect cannot be attained if the dependent variable measure is unreliable
- Conclusion
  - We do not get reliability from a single study
  - But if a population of studies give us the same results, then our dependent variable is reliable

Validity of the Dependent Variable

- What is the evidence that the dependent variable actually measures the construct we want it to measure?
  - Example: In an experiment on learning, does latency of response really represent degree of learning?

Guidelines for Achieving Construct Validity

- Clear definition of the construct
- Then
  - Obtain convergent and divergent data to provide additional verification that the construct has been measured

Guidelines for Achieving Construct Validity

- Convergent data
  - Data indicating agreement in several measures of the same construct
  - Example
    - High correlation between 2 different measures of a phenomenon (say, depression) by two different researchers is an indication that the two dependent variables are measuring the same thing
- Divergent data
  - Data indicating that the dependent variable is not related to conceptually different constructs
  - Example
    - A measure for depression should not correlate with a measure of intelligence because depression is a concept that is conceptually different from intelligence

Number of Dependent Variables

- Normally, when more than one dependent variable could be used, the scientist selects only one, and proceeds with the experiment
- But if the field is not fully advanced, there may be need to consider more than one dependent variable
  - Correlation between the dependent variables must be investigated
    - High correlation means that they are the same measure, and all but one of the dependent variables can be dropped
    - Lack of correlation may be due to unreliability of the measures, or the fact that they are not measuring the same aspect of the construct under study